Chandler Squires

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Contact Information

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Current Role

Postdoctoral Research Associate
Machine Learning Department, Carnegie Mellon University

Research Supervisor: Pradeep Ravikumar

Research Interests

Theoretical foundations and methodology (machine learning/statistics): Causality, representation learning, experimental design, decision-centric learning, neurosymbolic AI.

Scientific and engineering applications: Cellular biology, drug discovery, healthcare, manufacturing, mechanistic interpretability.

Education

Ph.D., Electrical Engineering and Computer Science **February 2025**
Massachusetts Institute of Technology, Cambridge, MA, USA

Thesis Advisors: David Sontag, Caroline Uhler

Thesis Title: Causal Foundations for Pragmatic Decision-Making (link)

M.Eng., Electrical Engineering and Computer Science **September 2019**
Massachusetts Institute of Technology, Cambridge, MA, USA

Thesis Advisor: Caroline Uhler

B.S., Electrical Engineering and Computer Science **June 2018**
Massachusetts Institute of Technology, Cambridge, MA, USA

Grants

DARPA Advanced Research Concepts (ARC) Critical Orientation Of Mathematics To Produce Advancements In Science And Security (COMPASS) **Sep. 2025–Aug. 2026**

Contextual Decision-Making Via Deep Preemptive Exploration

NSF, Robust Intelligence

Under development

Selected Publications

1. **Squires, C.**, Ravikumar, P., *A Unifying Framework for Unsupervised Concept Extraction*, AISTATS 2026.
2. Varıcı, B.*, **Squires, C.***, Ravikumar, P., *Causal Representation Learning*, Neurosymbolic AI: Foundations and Applications (2026).
3. **Squires, C.***, Choo, D.*, Bhattacharyya, A., Sontag, D., *Probably Approximately Correct High-Dimensional Causal Effect Estimation Given a Valid Adjustment Set*, CLeaR 2025.
4. **Squires, C.***, Seigal, A.*, Bhate, S., Uhler, C., *Linear Causal Disentanglement via Interventions*, ICML 2023.
5. **Squires, C.**, Uhler, C., *Causal Structure Learning: A Combinatorial Perspective*, JoFCM 2022.

Teaching Experience

Carnegie Mellon University (Guest Lectures)

1. CMU 10708, Probabilistic Graphical Models *(Instructor: Andrej Risteski)* **S. 2026**
2. CMU 10747, Neurosymbolic AI *(Instructor: Pradeep Ravikumar)* **F. 2025**
3. CMU 10708, Probabilistic Graphical Models *(Instructor: Andrej Risteski)* **S. 2025**
4. CMU 10716, Advanced ML *(Instructor: Pradeep Ravikumar)* **S. 2025**

5. CMU 10741, Representation Learning *(Instructor: Pradeep Ravikumar)* **F. 2024**

Massachusetts Institute of Technology (Instructor)

1. 6.S091, Causality (*Co-instructors: Katie Matton & Jiaqi Zhang*, link) **Jan. 2024**
2. 6.S091, Causality (link to lecture notes and recordings) **Jan. 2023**

Massachusetts Institute of Technology (Teaching Assistant)

1. 6.437, Inference and Information *(Instructors: Gregory Wornell & Polina Golland)* **S. 2019**
2. 6.438, Algorithms for Inference *(Instructor: Guy Bresler)* **F. 2018**

Mentorship

- **CMU PhD Students:** Juyong Kim, *2024–2025 (now ML engineer at Waymo)*; Soheun Yi, *2025–present*; Duncan Soiffer, *2025–present*; Hastyn Doshi, *2026–present*
- **CMU Master's Students:** Alex Guan, *2025–present*; Avi Amalanshu, *2025–present*; Yizhou Li, *2025–present*; Eddie Gao, *2025–present*
- **CMU Undergraduates:** Faraz Mizra, *2024–2025*; Leo Li, *2025–present*
- **CMU Visiting Students:** Annika Viswesh, *2025–present (visiting undergraduate from Caltech)*; Allan Pais, *2025–present (research engineer)*; Vansh Ramani, *2025–2026 (visiting undergraduate from IIT Delhi)*; Jianing Zhu, *2025–2026 (visiting PhD student from HKU)*
- **MIT Undergraduates:** Ryan Welch, *2023–2024 (now CS PhD at Stanford)*; Michael Truell, *2021–2023 (now CEO at Cursor)*; Sathwick Karnik, *2020–2022 (now PhD at Stanford)*; Joshua Amaniampong, *2021–2022 (now researcher at Citadel)*
- **MIT Master's Students:** Cathy Cai, *2023–2024 (now EECS PhD at MIT)*; Eshaan Nichani, *2020–2021 (now ECE PhD at Princeton)*; Neha Prasad, *2020–2021 (now ML engineer at Rillet)*; Annie Yun, *2020–2021 (now developer at HRT)*
- **MIT Visiting Students:** Álvaro Ribot, *2022–2023 (now Applied Math PhD at Harvard)*

Invited Talks

1. NeurIPS 2026 Workshop: Interpretability as a Science (link) **Scheduled: Dec. 2026**
   Title TBD
2. INFORMS 2026 Session: Pragmatic Causal Inference (link) **Nov. 2026**
   "Causal representation learning: a pragmatic viewpoint"
3. U. of Toronto: Mathematics of Machine Learning Seminar (link) **Sep. 2026**
   "Structured generalization: From identifiability theory to practical limitations"
4. U. of Pittsburgh: Department of Biomedical Informatics Colloquium (link) **Oct. 2025**
   "Causal representation learning for biology"
5. Bellairs Workshop on Causality: Causality in the Era of Foundation Models (link) **Feb. 2025**
   "An algebraic view of identifiability"
6. INFORMS 2024 Session: Causal Inference (link) **Oct. 2024**
   "Decision-centric causal structure learning: An algorithm for data-driven covariate adjustment"
7. NeurIPS 2023 Workshop: Causal Representation Learning (link) **Dec. 2023**
   "Causal imputation and causal disentanglement"
8. Valence Labs: Molecule Modeling and Drug Discovery (M_2D_2) **Aug. 2023**
   "Identifiability of causal models and applications to Perturb-seq data" (recording)
9. SIAM OP: Applications of Optimization for Causal Structure Learning (link) **Jun. 2023**
   "Linear causal disentanglement via interventions"
10. Colloquium on When Causal Inference meets Statistical Analysis (link) **Apr. 2023**
    "Beyond ICA: Causal disentanglement via interventions"

11. ICML 2022 Workshop: Principles of Distribution Shift (link) **Jul. 2022**
    "Causal structure learning with unknown mechanism shifts"
12. IMS 2022 Annual Meeting: Graphical Models Contributed Session (link) **Jun. 2022**
    "Causal structure discovery between clusters of nodes induced by latent factors"
13. First Interdisciplinary Workshop on Interactive Causal Learning **Jun. 2022**
    "The query complexity of verifying a causal graph from single-node interventions"
14. Simons Institute Semester on Causality: Causality Boot Camp (link) **Jan. 2022**
    "Introduction to experimental design for causal discovery" (recording)
15. U. of Amsterdam: AI4Science Colloquium (link) **May 2021**
    "Predicting gene expression across cell types and drugs" (recording)

Invited Panels

1. AISTATS 2026 Workshop: Causality in the Age of AI Scaling (link) **May 2026**
2. Emory University AI.Humanity Conference (link, recording) **Mar. 2026**

Professional Service

Assistant organizer, Identifiable Representation Learning Workshop at BIRS. (link) **2026**
Worked closely with Burak Varıcı to assemble a team of organizers (Aapo Hyvärinen, Dhanya Sridhar, Pradeep Ravikumar, and Soledad Vilar), submit a proposal, and organize a 35-person invite-only workshop at the Banff International Research Station, including talks, open problems sessions, a mentorship roundtable, and social events.

Workflow chair, Conference on Uncertainty in Artificial Intelligence (UAI). **2026**
Worked closely with Danielle Tsao and the program chairs (Daniel Malinsky and Emilija Perković) on the OpenReview workflow, from reviewer/AC recruitment through the reviewing and camera-ready stages.

Publication chair, Causal Learning and Reasoning (CLeaR). **2025 & 2026**
Prepared style files and final manuscripts for publication in the Proceedings of Machine Learning Research (PMLR).

Area chair, Conference on Neural Information Processing Systems (NeurIPS). **2025 & 2026**

Organizer, Causal Representation Learning Reading Group (link) **2021–2022**
Organized a bi-weekly reading group with members from over 20 universities.

Reviewer (Journals), JMLR, JOCI, TMLR, Bernoulli, AOS. **2022–present**

Reviewer (Conferences), NeurIPS, ICML, UAI, AISTATS, NeuS, CLeaR. **2019–present**
Awards: UAI 2021 Top Reviewer (top 10%), ICML 2026 Gold Reviewer (top 25%)

Institutional Service

Communication coach, MIT EECS Communication Lab **2021–2024**
Provided >100 hours of one-on-one feedback to EECS students and postdocs, focusing on technical communications such as writing, oral presentations, posters, and fellowship applications.
Ran workshops on scientific writing and presentation skills for EECS graduate students and developed a new workshop titled *"Effective Communication in Collaborative Research"*.

PhD Admissions Committee, EECS, MIT **2021**
Reviewed applications in the first round of the AI+D admissions process.

Tea talk committee member, LIDS, MIT **2020–2021**
Organized a series of weekly student talks within the LIDS community.

PhD Visit Days Team Leader, EECS, MIT **2020**

Prepared welcome materials representing ML/NLP/AI/robotics communities.

Treasurer, IDSS Student Council, MIT **2019–2020**

Tracked the council budget, approved reimbursements, and allocated funding to student-led academic and social events.

Social committee member, LIDS, MIT **2019–2020**

Organized weekly social gatherings for the LIDS community and hosted special events such as barbecues and game nights.

Refereed Publications

1. **Squires, C.**, Xie, Y., Stepputtis, S., Sycara, K., Ravikumar, P., *Position: Neurosymbolic AI is a Strong Technical Foundation for Trustworthy, Deployable AI by Design*, NeurIPS 2026 Position Paper Track (accepted).
2. Yi, S., Lu, Y., **Squires, C.**, Ravikumar, P., *Concept Modulation Models: A Unified Framework for Identifiability and Extrapolation*, NeurIPS 2026 (accepted) [arXiv].
3. Zhu, J., Li, Z., **Squires, C.**, Wang, Q., Han. B, Ravikumar, P., *On the Fragility of Latent Knowledge: Layerwise Influence under Unlearning in Large Language Models*, NeurIPS 2026 (accepted).
4. **Squires, C.**, Truell, M., Hütter, J.C., Zwiernik, P., Uhler, C., *Maximum Likelihood Estimation for Brownian Motion Tree Models Based on One Sample*, Bernoulli 2026.
5. Guan, Y., **Squires, C.**, Hu, T., Ravikumar, P., *Context-Informed Ship Trajectory Prediction via Conditional Attention*, IEEE FUSION 2026 [arXiv].
6. **Squires, C.**, Ravikumar, P., *A Unifying Framework for Unsupervised Concept Extraction*, AISTATS 2026 [arXiv].
7. Varıcı, B.*, **Squires, C.***, Ravikumar, P., *Causal Representation Learning*, Neurosymbolic AI: Foundations and Applications (2026).
8. Kim, J., **Squires, C.**, Ravikumar, P., *Knowledge-Enriched Machine Learning for Tabular Data*, NeuS 2025.
9. Brouillard, P., **Squires, C.**, Wahl, J., Kording, K., Sachs, K., Drouin, A., Sridhar, D., *The Landscape of Causal Discovery Data: Grounding Causal Discovery in Real-World Applications*, CLeaR 2025 [arXiv].
10. **Squires, C.***, Choo, D.*, Bhattacharyya, A., Sontag, D., *Probably Approximately Correct High-Dimensional Causal Effect Estimation Given a Valid Adjustment Set*, CLeaR 2025 [arXiv].
11. Mazaheri, B., **Squires, C.**, Uhler, C., *Synthetic Potential Outcomes and Causal Mixture Identifiability*, AISTATS 2025 [arXiv].
12. Ribot, Á., **Squires, C.**, Uhler, C., *Causal Imputation for Counterfactual SCMs: Bridging Graphs and Latent Factor Models*, CLeaR 2024 [arXiv].
13. Zhang, J., Cammarata, L., **Squires, C.**, Sapsis, T., Uhler, C., *Active Learning for Optimal Intervention Design in Causal Models*, Nature Machine Intelligence 2023 [arXiv].
14. Sturma, N., **Squires, C.**, Drton, M., Uhler, C., *Unpaired Multi-Domain Causal Representation Learning*, NeurIPS 2023 [arXiv].
15. Zhang, J., **Squires, C.**, Greenewald, K., Srivastava, A., Shanmugam, K., Uhler, C., *Identifiability Guarantees for Causal Disentanglement from Soft Interventions*, NeurIPS 2023 [arXiv].
16. Agrawal, R., **Squires, C.**, Prasad, N., Uhler, C., *The DeCAMFounder: Non-Linear Causal Discovery in the Presence of Hidden Variables*, JRSS-B 2023 [arXiv].
17. **Squires, C.***, Seigal, A.*, Bhate, S., Uhler, C., *Linear Causal Disentanglement via Interventions*, ICML 2023 [arXiv].
18. **Squires, C.***, Yun, A.*, Nichani, E., Agrawal, R., Uhler, C., *Causal Structure Discovery between Clusters of Nodes Induced by Latent Factors*, CLeaR 2022 [arXiv].

19. **Squires, C.***, Shen, D.*, Agarwal, A., Shah, D., Uhler, C., *Causal Imputation via Synthetic Interventions*, CLeaR 2022 [arXiv].
20. **Squires, C.**, Uhler, C., *Causal Structure Learning: A Combinatorial Perspective*, JoFCM 2022 [arXiv].
21. Zhang, J., **Squires, C.**, Uhler, C., *Matching a Desired Causal State via Shift Interventions*, NeurIPS 2021 [arXiv].
22. Belyaeva, A., **Squires, C.**, Uhler, C., *DCI: Learning Causal Differences between Gene Regulatory Networks*, Bioinformatics 2021 [pdf].
23. Belyaeva, A., Cammarata, L., Radhakrishnan, A., **Squires, C.**, Yang, K., Shivashankar, G.V., Uhler, C., *Causal Network Models of SARS-CoV-2 Expression and Aging to Identify Candidates for Drug Repurposing*, Nature Communications 2021 [arXiv].
24. **Squires, C.**, Magliacane, S., Greenewald, K., Katz, D., Kocaoglu, M., Shanmugam, K., *Active Structure Learning of Causal DAGs via Directed Clique Trees*, NeurIPS 2020 [arXiv].
25. **Squires, C.**, Wang, Y., Uhler, C., *Permutation-Based Causal Structure Learning with Unknown Intervention Targets*, UAI 2020 [arXiv].
26. Bernstein, D.*, Saeed, B.*, **Squires, C.***, Uhler, C., *Ordering-Based Causal Structure Learning in the Presence of Latent Variables*, AISTATS 2020 [arXiv].
27. Katz, D., Shanmugam, K., **Squires, C.**, Uhler, C., *Size of Interventional Markov Equivalence Classes in Random DAG Models*, AISTATS 2019 [arXiv].
28. Agrawal, R., **Squires, C.**, Yang, K., Shanmugam, K., Uhler, C., *ABCD-Strategy: Budgeted Experimental Design for Targeted Causal Structure Discovery*, AISTATS 2019 [arXiv].
29. Wang, Y., **Squires, C.**, Belyaeva, A., Uhler, C., *Direct Estimation of Differences in Causal Graphs*, NeurIPS 2018 [arXiv].

Preprints

1. Soiffer, D., **Squires, C.**, Guan, Y., Hartford, J., Ravikumar, P., *Catastrophic Compositional Generation: Why Vanilla Diffusion Models Fail to Extrapolate*, 2026 [arXiv].
2. Amalanshu, A., **Squires, C.**, Lu, Y., Ravikumar, P., *Estimating Implicit Expected Log Density Ratios*, 2026 [arXiv].
3. **Squires, C.**, Amalanshu, A., Li, L., Lu, T., Ravikumar, P., *Deep Preemptive Exploration*, 2026.

Industry Experience

Summer Researcher, Microsoft Research, Redmond, WA, USA **2021 (Jun.–Aug.)**

Developed a contrastive learning framework for improved multivariate time-series prediction in settings with causal relationships between variables.

Summer Researcher, IBM, Cambridge, MA, USA **2019 (Jun.–Aug.)**

Developed a theoretical characterization of optimal experimental design strategies for learning causal graphical models.

Data Science Intern, nference, Cambridge, MA, USA **2018 (Jan.–Aug.)**

Led both frontend and backend development for two new apps aimed at protein annotation and alignment and patient segmentation; analyzed custom statistical models of protein sequences.